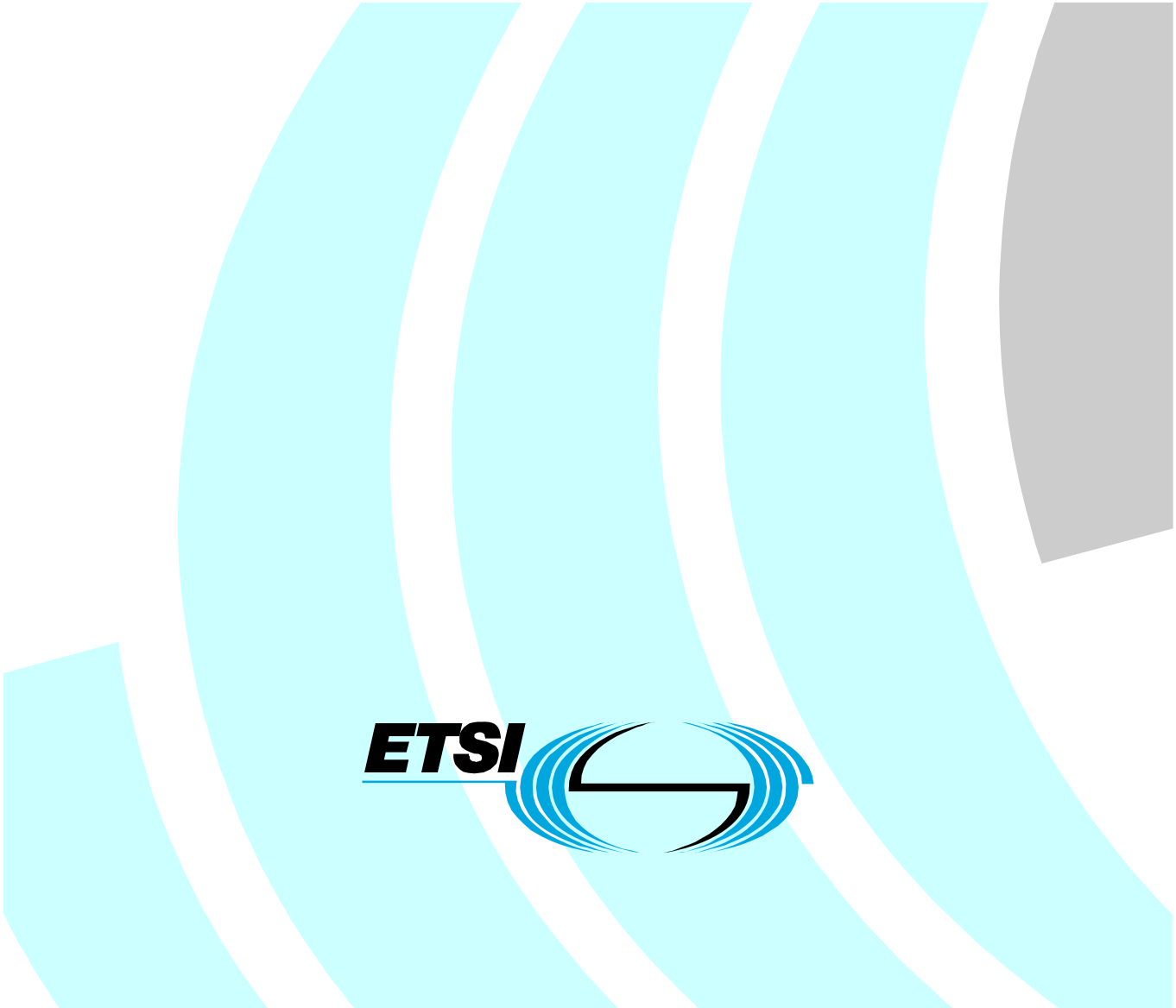
ETSI EN 301 908-12 V4.2.1 (2010-03)

Harmonized European Standard (Telecommunications series)

Electromagnetic compatibility and Radio spectrum Matters (ERM); Base Stations (BS), Repeaters and User Equipment (UE) for IMT-2000 Third-Generation cellular networks; Part 12: Harmonized EN for IMT-2000, CDMA Multi-Carrier (cdma2000) (Repeaters) covering the essential requirements of article 3.2 of the R&TTE Directive



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Contents

Intellectual Property Rights 5
Foreword 5
Introduction 6
1 Scope 7
2 References 7
2.1 Normative references 8
2.2 Informative references 8
3 Definitions, symbols and abbreviations 8
3.1 Definitions 8
3.2 Symbols 10
3.3 Abbreviations 11
4 Technical requirements specifications 11
4.1 Environmental profile 11
4.2 Conformance requirements 11
4.2.1 Introduction 11
4.2.2 Spectrum emissions mask 12
4.2.2.1 Definition 12
4.2.2.2 Limits 12
4.2.2.3 Conformance 13
4.2.3 Spurious emissions 13
4.2.3.1 Definition 13
4.2.3.2 Limits 13
4.2.3.2.1 Spurious emissions 13
4.2.3.3 Conformance 16
4.2.4 Maximum output power 16
4.2.4.1 Definition 16
4.2.4.2 Limits 16
4.2.4.3 Conformance 16
4.2.5 Input intermodulation 16
4.2.5.1 Definition 16
4.2.5.2 Limits 17
4.2.5.3 Conformance 17
4.2.6 Out-of-band gain 17
4.2.6.1 Definition 17
4.2.6.2 Limits 17
4.2.6.3 Conformance 18
4.2.7 Output intermodulation 18
4.2.7.1 Definition 18
4.2.7.2 Limits 18
4.2.7.3 Conformance 18
5 Testing for compliance with technical requirements 19
5.1 Conditions for testing 19
5.1.1 Introduction 19
5.1.2 Standard equipment under test 19
5.2 Interpretation of the measurement results 19
5.3 Essential radio test suites 19
5.3.1 Spectrum emission mask 19
5.3.1.1 Initial conditions 19
5.3.1.2 Procedures 20
5.3.2 Spurious emissions 20
5.3.2.1 Initial conditions 20
5.3.2.2 Procedure 20
5.3.3 Maximum output power 20

5.3.3.1 Initial conditions 20
5.3.3.2 Procedure 20
5.3.4 Input intermodulation 21
5.3.4.1 Initial conditions 21
5.3.4.2 Procedures 21
5.3.5 Out-of-band gain 22
5.3.5.1 Initial conditions 22
5.3.5.2 Procedure 22
5.3.6 Output intermodulation 23
5.3.6.1 Initial conditions 23
5.3.6.2 Procedure 23

Annex A (normative): HS Requirements and conformance Test specifications Table (HS-RTT) 24

Annex B (normative): Repeater configurations 26

B.1 Power supply 26

B.2 Power supply options 26

B.3 Ancillary equipment 26

Annex C (informative): Environmental profile specification 27

Annex D (informative): The EN title in the official languages 28

Annex E (informative): Bibliography 29

History 30

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Foreword

This Harmonized European Standard (Telecommunications series) has been produced by ETSI Technical Committee Electromagnetic compatibility and Radio spectrum Matters (ERM).

The present document has been produced by ETSI in response to a mandate from the European Commission issued under Council Directive 98/34/EC [i.1] (as amended) laying down a procedure for the provision of information in the field of technical standards and regulations.

The present document is intended to become a Harmonized Standard, the reference of which will be published in the Official Journal of the European Communities referencing the Directive 1999/5/EC [i.2] of the European Parliament and of the Council of 9 March 1999 on radio equipment and telecommunications terminal equipment and the mutual recognition of their conformity ("the R&TTE Directive").

Technical specifications relevant to Directive 1999/5/EC [i.2] are given in annex A.

The present document is part 12 of a multi-part deliverable covering the Base Stations (BS), Repeaters and User Equipment (UE) for IMT-2000 Third-Generation cellular networks, as identified below:

Part 1: "Harmonized EN for IMT-2000, introduction and common requirements, covering the essential requirements of article 3.2 of the R&TTE Directive";

Part 2: "Harmonized EN for IMT-2000, CDMA Direct Spread (UTRA FDD and E-UTRA FDD) (UE) covering the essential requirements of article 3.2 of the R&TTE Directive";

Part 3: "Harmonized EN for IMT-2000, CDMA Direct Spread (UTRA FDD and E-UTRA FDD) (BS) covering the essential requirements of article 3.2 of the R&TTE Directive";

Part 4: "Harmonized EN for IMT-2000, CDMA Multi-Carrier (cdma2000) and Evolved CDMA Multi-Carrier Ultra Mobile Broadband (UMB) (UE) covering the essential requirements of article 3.2 of the R&TTE Directive";

Part 5: "Harmonized EN for IMT-2000, CDMA Multi-Carrier (cdma2000) and Evolved CDMA Multi-Carrier Ultra Mobile Broadband (UMB) (BS) covering the essential requirements of article 3.2 of the R&TTE Directive";

Part 6: "Harmonized EN for IMT-2000, CDMA TDD (UTRA TDD and E-UTRA TDD) (UE) covering the essential requirements of article 3.2 of the R&TTE Directive";

Part 7: "Harmonized EN for IMT-2000, CDMA TDD (UTRA TDD and E-UTRA TDD) (BS) covering the essential requirements of article 3.2 of the R&TTE Directive";

Part 8: "Harmonized EN for IMT-2000, TDMA Single-Carrier (UWC 136) (UE) covering essential requirements of article 3.2 of the R&TTE Directive";

Part 9: "Harmonized EN for IMT-2000, TDMA Single-Carrier (UWC 136) (BS) covering essential requirements of article 3.2 of the R&TTE Directive";

Part 10: "Harmonized EN for IMT-2000, FDMA/TDMA (DECT) covering essential requirements of article 3.2 of the R&TTE Directive";

Part 11: "Harmonized EN for IMT-2000, CDMA Direct Spread (UTRA FDD and E-UTRA FDD) (Repeaters) covering the essential requirements of article 3.2 of the R&TTE Directive";

Part 12: "Harmonized EN for IMT-2000, CDMA Multi-Carrier (cdma2000) (Repeaters) covering the essential requirements of article 3.2 of the R&TTE Directive";

Part 13: "Harmonized EN for IMT-2000, Evolved Universal Terrestrial Radio Access (E-UTRA) (UE) covering the essential requirements of article 3.2 of the R&TTE Directive";

Part 14: "Harmonized EN for IMT-2000, Evolved Universal Terrestrial Radio Access (E-UTRA) (BS) covering the essential requirements of article 3.2 of the R&TTE Directive";

Part 15: "Harmonized EN for IMT-2000, Evolved Universal Terrestrial Radio Access (E-UTRA) (FDD Repeaters) covering the essential requirements of article 3.2 of the R&TTE Directive";

Part 16: "Harmonized EN for IMT-2000, Evolved CDMA Multi-Carrier Ultra Mobile Broadband (UMB) (UE) covering the essential requirements of article 3.2 of the R&TTE Directive";

Part 17: "Harmonized EN for IMT-2000, Evolved CDMA Multi-Carrier Ultra Mobile Broadband (UMB) (BS) covering the essential requirements of article 3.2 of the R&TTE Directive".

National transposition dates	
Date of adoption of this EN:	19 February 2010
Date of latest announcement of this EN (doa):	31 May 2010
Date of latest publication of new National Standard or endorsement of this EN (dop/e):	30 November 2010
Date of withdrawal of any conflicting National Standard (dow):	30 November 2011

Introduction

The present document is part of a set of standards developed by ETSI and is designed to fit in a modular structure to cover all radio and telecommunications terminal equipment within the scope of the R&TTE Directive [i.2]. The modular structure is shown in EG 201 399 [i.3].

1 Scope

The present document applies to the following radio equipment types:

- Repeaters for IMT-2000 CDMA multi-carrier (cdma2000).

These radio equipment types are capable of operating in all or any part of the frequency bands given in table 1-1.

Table 1-1: CDMA multi-carrier Repeater operating bands

Band Class (BC)	Direction of transmission	CDMA multi-carrier Repeater operating bands
6	Forward link (DL)	2 110 MHz to 2 170 MHz
	Reverse link (UL)	1 920 MHz to 1 980 MHz
8	Forward link (DL)	1 805 MHz to 1 880 MHz
	Reverse link (UL)	1 710 MHz to 1 785 MHz
9	Forward link (DL)	925 MHz to 960 MHz
	Reverse link (UL)	880 MHz to 915 MHz
13	Forward link (DL)	2 620 MHz to 2 690 MHz
	Reverse link (UL)	2 500 MHz to 2 570 MHz

Repeaters for IMT-2000 CDMA multi-carrier (cdma2000) may support:

1) operation in cdma2000 spread spectrum systems as defined in 3GPP2 C.S0002-D [2], referred to herein as operation in Type 1 cdma2000 systems; or

2) operation in cdma2000 High Rate Packet Data Systems as defined in TIA-856 [3], referred to herein in Type 2 cdma2000 systems.

The present document is intended to cover the provisions of Directive 1999/5/EC [i.2] (R&TTE Directive) article 3.2, which states that "[…] radio equipment shall be so constructed that it effectively uses the spectrum allocated to terrestrial/space radio communications and orbital resources so as to avoid harmful interference".

In addition to the present document, other ENs that specify technical requirements in respect of essential requirements under other parts of article 3 of the R&TTE Directive [i.2] may apply to equipment within the scope of the present document.

NOTE: A list of such ENs is included on the web site http://www.newapproach.org/.

2 References

References are either specific (identified by date of publication and/or edition number or version number) or non-specific.

- For a specific reference, subsequent revisions do not apply.
- Non-specific reference may be made only to a complete document or a part thereof and only in the following cases:
  - if it is accepted that it will be possible to use all future changes of the referenced document for the purposes of the referring document;
  - for informative references.

Referenced documents which are not found to be publicly available in the expected location might be found at http://docbox.etsi.org/Reference.

NOTE: While any hyperlinks included in this clause were valid at the time of publication ETSI cannot guarantee their long term validity.

2.1 Normative references

The following referenced documents are indispensable for the application of the present document. For dated references, only the edition cited applies. For non-specific references, the latest edition of the referenced document (including any amendments) applies.

[1] ANSI/TIA-97-F-1 (June 2006): "Recommended Minimum Performance Standards for cdma2000 Spread Spectrum Base Stations - Addendum 1".

[2] 3GPP2 C.S0002-D V2.0 (September 2005): "Physical Layer Standard for cdma2000 Spread Spectrum Systems Revision D".

NOTE: Available at: http://www.3gpp2.org/Public_html/Specs/C.S0002-D_v2.0_051006.pdf.

[3] TIA-856-B (October 2007): "cdma2000 High Rate Packet Data Air Interface Specification".

[4] 3GPP2 C.S0032-B V1.0 (May 2008): "Recommended Minimum Performance Standards for cdma2000 High Rate Packet Data Access Network".

NOTE: Available at: http://www.3gpp2.org/Public_html/specs/C.S0032-B_v1.0_080519.pdf.

[5] TIA-1037 (May 2008): "Recommended Minimum Performance Standards for cdma2000 Spread Spectrum Repeaters".

[6] TIA-1030-B (December 2006): "Band Class Specification for cdma2000 Spread Spectrum Systems".

[7] ETSI EN 301 908-1 (V4.1.2): "Electromagnetic compatibility and Radio spectrum Matters (ERM); Base Stations (BS), Repeaters and User Equipment (UE) for IMT-2000 Third-Generation cellular networks; Part 1: Harmonized EN for IMT-2000, introduction and common requirements, covering the essential requirements of article 3.2 of the R&TTE Directive".

2.2 Informative references

The following referenced documents are not essential to the use of the present document but they assist the user with regard to a particular subject area. For non-specific references, the latest version of the referenced document (including any amendments) applies.

[i.1] Directive 98/34/EC of the European Parliament and of the Council of 22 June 1998 laying down a procedure for the provision of information in the field of technical standards and regulations. .

[i.2] Directive 1999/5/EC of the European Parliament and of the Council of 9 March 1999 on radio equipment and telecommunications terminal equipment and the mutual recognition of their conformity (R&TTE Directive).

[i.3] ETSI EG 201 399 (V2.1.1): "Electromagnetic compatibility and Radio spectrum Matters (ERM); A guide to the production of candidate Harmonized Standards for application under the R&TTE Directive".

3 Definitions, symbols and abbreviations

3.1 Definitions

For the purposes of the present document, the following terms and definitions apply:

Band Class (BC): set of frequency channels and a numbering scheme for these channels

Base Station (BS): fixed station used for communicating with mobile stations

NOTE 1: Base stations for IMT-2000 CDMA multi-carrier (cdma2000) may support, operation in cdma2000 spread spectrum systems as defined in 3GPP2 C.S0002-D [2], referred to herein as operation in Type 1 cdma2000 system, or operation in cdma2000 High Rate Packet Data Systems as defined in TIA-856 [3], referred to herein as operation in Type 2 cdma2000 systems.

NOTE 2: Depending upon the context, the term Base Station may refer to a cell, a sector within a cell, an MSC, and access network or other part of the wireless system. See also MSC.

CDMA channel: set of channels transmitted from the Base Station and the mobile stations on a given frequency

CDMA channel number: 11-bit number corresponding to the centre of the CDMA frequency assignment

CDMA frequency assignment: 1,23 MHz segment of spectrum

NOTE: For band classes 6, 8 and 9, the channel is centred on one of the 50 kHz channels.

chip rate: rate of "chips" (modulated symbols after spreading) per second

Code Division Multiple Access (CDMA): technique for spread-spectrum multiple-access digital communications that creates channels through the use of unique code sequences

donor coupling loss: coupling loss between the Repeater and the donor Base Station

DownLink (DL): signal path where Base Station or Repeater transmits and the mobile receives

NOTE: Also referred to as the forward link.

environmental profile: range of environmental conditions under which equipment within the scope of the present document is required to comply with the provisions of the present document

forward CDMA channel: CDMA channel from a Base Station or Repeater to mobile stations

NOTE: The forward CDMA channel contains one or more code channels that are transmitted on a CDMA frequency assignment using a particular pilot PN offset.

High Rate Packet Data (HRPD): CDMA technique optimized for data communications in Type 2 cdma2000 systems

maximum output power (Pmax): mean power level per carrier measured at the antenna connector of the Repeater in specified reference condition

mean power: total calorimetric power measured in a specified bandwidth at the antenna connector

mobile station: station intended to be used while in motion or during halts at unspecified points

NOTE: Mobile stations include portable units (e.g. hand-held personal units) and units installed in vehicles and HRPD access terminals.

pass band: frequency range that the Repeater operates in with operational configuration

NOTE 1: This frequency range can correspond to one or several consecutive nominal 5 MHz channels. If they are not consecutive each subset of channels is considered as an individual pass band.

NOTE 2: The Repeater can have one or several pass bands.

radio configuration: set of forward traffic channel and reverse traffic channel transmission formats that are characterized by physical layer parameters such as transmission rates, modulation characteristics, and spreading rate

NOTE: Radio configurations are defined in 3GPP2 C.S0002-D [2], clauses 2.1.3 and 3.1.3.

Resolution BandWidth (RBW): measured in Hz and represents the frequency over which power is integrated in a spectrum analyser to display the amplitude at the centre of the integration frequency range

repeater: device that receives, amplifies and transmits the radiated or conducted RF carrier both in the down-link direction (from the Base Station to the mobile area) and in the up-link direction (from the mobile to the Base Station)

NOTE: A repeater can be a device that receives, amplifies and transmits one or more radiated or conducted CDMA channel(s) both in the down-link direction (from the Base Station to the mobile area) and in the up-link direction (from the mobile to the Base Station).

reverse CDMA channel: CDMA channel from the mobile station to the Base Station

NOTE: From the Base Station's perspective, the reverse CDMA channel is the sum of all mobile station transmissions on a CDMA frequency assignment.

RF carrier: direct-sequence spread RF channel

NOTE: For the forward CDMA channel, the number of RF carriers is 1 for spreading rate 1 and 3 for spreading rate 3; for the reverse CDMA channel, there is one RF carrier.

spreading rate: PN chip rate of the forward CDMA channel or the reverse CDMA channel, defined as a multiple of 1,2288 Mcps

spreading rate 1: often referred to as "1X"

NOTE 1: A spreading rate 1 forward CDMA channel uses a single direct-sequence spread carrier with a chip rate of 1,2288 Mcps.

NOTE 2: A spreading rate 1 reverse CDMA channel uses a single direct-sequence spread carrier with a chip rate of 1,2288 Mcps.

spreading rate 3: often referred to as "3X"

NOTE 1: A spreading rate 3 forward CDMA channel uses three direct-sequence spread carriers (see multiple-carrier forward channel) each with a chip rate of 1,2288 Mcps.

NOTE 2: A spreading rate 3 reverse CDMA channel uses a single direct-sequence spread carrier with a chip rate of 3,6864 Mcps.

UpLink (UL): signal path where the mobile or Repeater transmits and the Base Station receives

NOTE: Also referred to as the reverse link.

User Equipment (UE): mobile station supporting operation in cdma2000 spread spectrum systems

NOTE: See 3GPP2 C.S0002-D [2], referred to herein as operation in Type 1 cdma2000 system; access terminal supporting operation in cdma2000 High Rate Packet Data Systems as defined in TIA-856 [3], referred to herein as operation in Type 2 cdma2000 system; and mobile station supporting operation in Type 1 and Type 2 cdma2000 systems.

3.2 Symbols

For the purposes of the present document, the following symbols apply:

Δf	frequency offset from the center frequency
Ω	Ohm
dBc	ratio (in dB) of the sideband power of a signal, measured in a given bandwidth at a given frequency offset from the centre frequency of the same signal, to the total inband power of the signal

NOTE: For CDMA, the total inband power of the signal is measured in a 1,23 MHz bandwidth around the centre frequency of the CDMA signal for a spreading rate 1 CDMA signal and in 3,69 MHz bandwidth around the centre frequency of the CDMA signal for a spreading rate 3 CDMA signal.

dBm	measure of power expressed in terms of its ratio (in dB) to 1 mW
dBm/Hz	measure of power spectral density

NOTE: The ratio, dBm/Hz, is the power in 1 Hz of bandwidth, where power is expressed in units of dBm.

GHz	GigaHertz (10^9 Hertz)
kHz	kiloHertz (10^3 Hertz)
MHz	MegaHertz (10^6 Hertz)
Pout	transmitter RF output power

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

BC	Band Class
BS	Base Station
CDMA	Code Division Multiple Access
CW	Continuous Wave (unmodulated signal)
DCS	Digital Cellular System
DL	DownLink
E-UTRA	Evolved Universal Terrestrial Radio Access
EMC	ElectroMagnetic Compatibility
GSM	Global System for Mobile Communications
HRPD	High Rate Packet Data
IMT	International Mobile Telecommunications
MSC	Mobile Switching Centre
PN	PseudoNoise
R&TTE	Radio and Telecommunications Terminal Equipment
RBW	Resolution BandWidth
RF	Radio Frequency
TDD	Time Division Duplex
UE	User Equipment
UL	UpLink
UTRA	Universal Terrestrial Radio Access

4 Technical requirements specifications

4.1 Environmental profile

The technical requirements of the present document apply under the environmental profile for operation of the equipment, which shall be declared by the supplier. The equipment shall comply with all the technical requirements of the present document at all times when operating within the boundary limits of the declared operational environmental profile.

For guidance on how a supplier can declare the environmental profile see annex C.

4.2 Conformance requirements

The requirements in the present document are based on the assumption that the operating band is shared between systems of the IMT-2000 family or systems having compatible characteristics.

4.2.1 Introduction

To meet the essential requirement under article 3.2 of Directive 1999/5/EC [i.2] (R&TTE Directive) for IMT-2000 Repeater equipment five essential parameters in addition to those in EN 301 908-1 [7] have been identified.
Table 4.2.1-1 provides a cross reference between these five essential parameters and the six corresponding technical requirements for equipment within the scope of the present document.

Table 4.2.1-1: Cross references

Essential parameter	Corresponding technical requirements
Spectrum emissions mask	4.2.2 Spectrum emissions mask
Conducted spurious emissions from the antenna connector	4.2.3 Spurious emissions
Accuracy of maximum output power	4.2.4 Maximum output power
Receiver immunity	4.2.5 Input intermodulation
	4.2.6 Out-of-band gain
Intermodulation attenuation of the output	4.2.7 Output intermodulation

The supplier shall declare operating band(s) for the Repeater. The technical requirements apply for the declared operating band(s) as outlined for each requirement. For a Repeater supporting more than one operating band, conformance testing for each technical requirement in clause 5 shall be performed independently for each operating band.

The technical requirements also apply to Repeater configurations described in annex B.

4.2.2 Spectrum emissions mask

4.2.2.1 Definition

Out-of-band emissions are emissions immediately outside the channel bandwidth resulting from the modulation process and non-linearity in the transmitter but excluding spurious emissions. This out-of-band emission limit is specified in terms of a spectrum emission mask for the transmitter.

4.2.2.2 Limits

The requirement shall be met by a Repeater's RF-signal output at maximum gain in the pass band of the Repeater, at levels that produce the maximum rated output power per channel, configured in accordance with the manufacturer's specification. Emissions shall not exceed the maximum level specified in tables 4.2.2.2-1 and 4.2.2.2-2 for the appropriate Repeater maximum output power.

If the pass band corresponds to two or more consecutive nominal carriers, the requirement shall be met with any combination of two CDMA modulated signals of equal power in the Repeaters pass band.

Table 4.2.2.2-1: Spectrum emission mask values, Band Class 9

For $\|\Delta f\|$ Within the Range	Applies to Multiple Carriers	Emission limit
750 kHz to 1,98 MHz	No	-45 dBc/30 kHz
1,98 MHz to 4,00 MHz	No	-60 dBc/30 kHz; Pout ≥ 33 dBm -27 dBm/30 kHz; 28 dBm ≤ Pout < 33 dBm -55 dBc/30 kHz; Pout < 28 dBm
NOTE: All frequencies in the measurement bandwidth shall satisfy the restrictions on $\|\Delta f\|$ where Δf = centre frequency - closer measurement edge frequency (f). The emissions requirements shall apply for all values of Δf regardless of whether the measurement frequency falls inside or outside of the band or block edge. Compliance with the -46 dBm/6,25 kHz limit is based on the use of measurement instrumentation such that the reading taken with any resolution bandwidth setting should be adjusted to indicate spectral power in a 6,25 kHz segment. For multiple-carrier testing, Δf is defined for positive Δf as the centre frequency of the highest carrier - closer measurement edge frequency (f) and for negative Δf as the centre frequency of the lowest carrier - closer measurement edge frequency (f).		

Table 4.2.2.2-2: Spectrum emission mask values, Band Class 6, 8 and 13

For $\|\Delta f\|$ Within the Range	Applies to Multiple Carriers	Emission limit
885 kHz to 1,25 MHz	No	-45 dBc/30 kHz
1,25 MHz to 1,98 MHz	No	More stringent of -45 dBc/30 kHz or -9 dBm/30 kHz
1,25 MHz to 1,45 MHz	Yes	-13 dBm/30 kHz
1,45 MHz to 2,25 MHz	Yes	$-[13 + 17 \times (\|\Delta f\| - 1{,}45\ \text{MHz})]$ dBm/30 kHz
1,98 MHz to 2,25 MHz	No	-55 dBc/30 kHz; Pout $\geq$ 33 dBm -22 dBm/30 kHz; 28 dBm $\leq$ Pout < 33 dBm -50 dBc/30 kHz; Pout < 28 dBm
2,25 MHz to 4,00 MHz	Yes	-13 dBm/1 MHz
NOTE: All frequencies in the measurement bandwidth shall satisfy the restrictions on $\|\Delta f\|$ where Δf = centre frequency - closer measurement edge frequency (f). The emissions requirements shall apply for all values of Δf regardless of whether the measurement frequency falls inside or outside of the band or block edge. For multiple-carrier testing, Δf is defined for positive Δf as the centre frequency of the highest carrier - closer measurement edge frequency (f) and for negative Δf as the centre frequency of the lowest carrier - closer measurement edge frequency (f).		

4.2.2.3 Conformance

Conformance tests described in clause 5.3.1 shall be carried out.

4.2.3 Spurious emissions

4.2.3.1 Definition

Spurious emissions are emissions which are caused by unwanted transmitter effects such as harmonics emission, parasitic emission, intermodulation products and frequency conversion products, but exclude out-of-band emissions. This is measured at the Repeater output port.

The requirements of clause 4.2.3.2 shall apply whatever the type of Repeater considered (one or several pass bands). It applies for all configurations foreseen by the manufacturer's specification.

4.2.3.2 Limits

4.2.3.2.1 Spurious emissions

The mean spurious emissions levels shall comply with the applicable band class limits contained in tables 4.2.3.2.1-1 to 4.2.3.2.1-5 for every specified test.

If the pass band corresponds to two or more consecutive nominal carriers, the requirement shall be met with any combination of two CDMA modulated signals of equal power in the Repeaters pass band.

When transmitting in Band Class 9, the mean spurious emissions shall be less than all of the limits specified in table 4.2.3.2.1-1 for both the Forward and Reverse signal paths.

Table 4.2.3.2.1-1: Band Class 9 transmitter spurious emission limits

For $\|\Delta f\|$ Within the Range	Applies to Multiple Carriers	Emission limit	
> 4,00 MHz	Yes	-36 dBm/1 kHz; -36 dBm/10 kHz; -30 dBm/1 MHz;	9 kHz < f < 150 kHz 150 kHz < f < 30 MHz 1 GHz < f < 12,75 GHz
4,00 MHz to 6,40 MHz	Yes	-36 dBm/1 kHz	30 MHz < f < 1 GHz
6,40 MHz to 16 MHz	Yes	-36 dBm/10 kHz	30 MHz < f < 1 GHz
> 16 MHz	Yes	-36 dBm/100 kHz	30 MHz < f < 1 GHz
NOTE: All frequencies in the measurement bandwidth shall satisfy the restrictions on $\|\Delta f\|$ where Δf = centre frequency - closer measurement edge frequency (f). The emissions requirements shall apply for all values of Δf regardless of whether the measurement frequency falls inside or outside of the band or block edge. Compliance with the -46 dBm/6,25 kHz limit is based on the use of measurement instrumentation such that the reading taken with any resolution bandwidth setting should be adjusted to indicate spectral power in a 6,25 kHz segment. For multiple-carrier testing, Δf is defined for positive Δf as the centre frequency of the highest carrier - closer measurement edge frequency (f) and for negative Δf as the centre frequency of the lowest carrier - closer measurement edge frequency (f).			

When transmitting in Band Class 6, 8 or 13 the mean spurious emissions shall be less than all of the limits specified in table 4.2.3.2.1-2 for both the Forward and Reverse signal paths.

Table 4.2.3.2.1-2: Band Class 6, 8 and 13 transmitter spurious emission limits

For $\|\Delta f\|$ Within the Range	Applies to Multiple Carriers	Emission limit	
> 4,00 MHz	Yes	-36 dBm/1 kHz -36 dBm/10 kHz -36 dBm/100 kHz	9 kHz < f < 150 kHz 150 kHz < f < 30 MHz 30 MHz < f < 1 GHz
4,00 MHz to 16,0 MHz	Yes	-30 dBm/30 kHz	1 GHz < f < 12,75 GHz
16,0 MHz to 19,2 MHz	Yes	-30 dBm/300 kHz	1 GHz < f < 12,5 GHz
> 19,2 MHz	Yes	-30 dBm/1 MHz	1 GHz < f < 12,75 GHz
NOTE: All frequencies in the measurement bandwidth shall satisfy the restrictions on $\|\Delta f\|$ where Δf = centre frequency - closer measurement edge frequency (f). The emissions requirements shall apply for all values of Δf regardless of whether the measurement frequency falls inside or outside of the band or block edge. For multiple-carrier testing, Δf is defined for positive Δf as the centre frequency of the highest carrier - closer measurement edge frequency (f) and for negative Δf as the centre frequency of the lowest carrier - closer measurement edge frequency (f).			

Mean spurious emissions shall also be less than the limits specified in table 4.2.3.2.1-3 for the forward link. For the reverse link spurious emissions shall also be less than the limits specified in tables 4.2.3.2.1-4 and 4.2.3.2.1-5.

Table 4.2.3.2.1-3: Additional forward link transmitter spurious emission limits

Measurement Frequency	Applies to Multiple Carriers	Emission limit	When Coverage Overlaps With	Applies to Operating Band
876 MHz to 915 MHz	No	-61 dBm/100 kHz	GSM900	BC 6, 8, 13
921 MHz to 960 MHz	Yes	-57 dBm/100 kHz	GSM900	BC 6, 8, 13
1 710 MHz to 1 785 MHz	No	-61 dBm/100 kHz	DCS1800	BC 6, 9, 13
1 805 MHz to 1 880 MHz	Yes	-47 dBm/100 kHz	DCS1800	BC 6, 9, 13
1 900 MHz to 1 920 MHz, 2 010 MHz to 2 025 MHz, and 2 570 MHz to 2 610 MHz	Yes	-52 dBm/1 MHz	UTRA-TDD	BC 6, 8, 9, 13
1 920 MHz to 1 980 MHz	No	-86 dBm/1 MHz	Always	BC 6
1 710 MHz to 1 785 MHz	No	-86 dBm/1 MHz	Always	BC 8
876 MHz to 915 MHz	No	-86 dBm/1 MHz	Always	BC 9
2 500 MHz to 2 570 MHz	No	-86 dBm/1 MHz	Always	BC 13

Table 4.2.3.2.1-4: Additional reverse link transmitter spurious emission limits

Measurement frequency	Emission limit	When Coverage Overlaps With	Applies to Operating Band
921 MHz to 925 MHz	-60 dBm/100 kHz	GSM900	BC 6, 8, 13
925 MHz to 935 MHz	-67 dBm/100 kHz	GSM900	BC 6, 8, 13
935 MHz to 960 MHz	-79 dBm/100 kHz	GSM900	BC 6, 8, 13
1 805 MHz to 1 880 MHz	-71 dBm/100 kHz	DCS1800	BC 6, 9, 13
2 110 MHz to 2 170 MHz	-60 dBm/1,23 MHz	IMT-2000 BC 6	BC 8, 9, 13
2 590 MHz to 2 620 MHz	-50 dBm/1,23 MHz	IMT-2000 BC 13 TDD	BC 13
2 620 MHz to 2 690 MHz	-60 dBm/1,23 MHz	IMT-2000 BC 13 FDD	BC 9, 13
NOTE: Measurements apply only when the measurement frequency is at least 5,625 MHz from the CDMA centre frequency. The measurements shall be made on frequencies which are integer multiples of 200 kHz. As exceptions, up to five measurements with a level up to the spurious emission limits in tables 4.2.3.2.1-1 and 4.2.3.2.1-2 are allowed.			

Table 4.2.3.2.1-5: Repeater spurious emissions limits in geographic coverage area of GSM900, DCS1800 and/or BS receiver for the reverse link direction of the Repeater

BS type operating in the same geographical area	Measurement Band	Maximum level	Measurement bandwidth
GSM900	876 MHz to 915 MHz	-47 dBm	100 kHz
DCS1800	1 710 MHz to 1 785 MHz	-47 dBm	100 kHz
IMT-2000 BC 6	1 920 MHz to 1 980 MHz	-47 dBm	100 kHz
IMT-2000 BC 8	1 710 MHz to 1 785 MHz	-47 dBm	100 kHz
IMT-2000 BC 9	880 MHz to 915 MHz	-47 dBm	100 kHz
IMT-2000 BC 13	2 500 MHz to 2 570 MHz	-47 dBm	100 kHz
NOTE: Measurements apply only when the measurement frequency is at least 12,5 MHz from the CDMA centre frequency. The measurements shall be made on frequencies which are integer multiples of 200 kHz. As exceptions, up to five measurements with a level up to the spurious emission limits in tables 4.2.3.2.1-1 and 4.2.3.2.1-2 are allowed.			

4.2.3.3 Conformance

Conformance tests described in clause 5.3.2 shall be carried out.

4.2.4 Maximum output power

4.2.4.1 Definition

Maximum output power, Pmax, of the Repeater is the mean power level per carrier measured at the antenna connector in specified reference condition.

4.2.4.2 Limits

When tested using the standard test environmental requirements as specified in clause 5.1 in TIA-1037 [5], the Repeater's mean output power shall remain within limits specified in table 4.2.4.2-1 relative to the manufacturer's rated mean output under every specified test.

Table 4.2.4.2-1: Repeater output power: normal conditions

Rated output power	Limit
P ≥ 43 dBm	+2,0 dB and -2,0 dB
39 dBm ≤ P < 43 dBm	+2,0 dB and -2,0 dB
31 dBm ≤ P < 39 dBm	+2,0 dB and -2,0 dB
P < 31 dBm	+3,0 dB and -3,0 dB

When tested using the extreme environmental test conditions specified in clause 4.1 in TIA-1037 [5], the Repeater's mean output power shall remain within limits specified in table 4.2.4.2-2 relative to the manufacturer's rated mean output power.

Table 4.2.4.2-2: Repeater output power: extreme conditions

Rated output power	Limit
P ≥ 43 dBm	+2,0 dB and -4,0 dB
39 dBm ≤ P < 43 dBm	+2,0 dB and -4,0 dB
31 dBm ≤ P < 39 dBm	+2,0 dB and -4,0 dB
P < 31 dBm	+3,5 dB and -5,5 dB

4.2.4.3 Conformance

Conformance tests described in clause 5.3.3 shall be carried out.

4.2.5 Input intermodulation

4.2.5.1 Definition

Input intermodulation spurious response attenuation is a measure of a Repeater's ability to rebroadcast an in-band signal in the presence of two interfering out-of-band CW signals at the input of the Repeater. For Repeaters specified by the manufacturer as not suitable for use as an Over the Air Repeater, this test only applies to the reverse link.

4.2.5.2 Limits

The intermodulation performance should be met when the following signals are applied to the Repeater.

Table 4.2.5.2-1: General input intermodulation requirement

Band Class	CW Generator Power at RF input port of Repeater	CW Generator Frequencies	Measurement bandwidth
9	-40 dBm	F1 - 900 kHz and F1 - 1 800 kHz	10 kHz
	-40 dBm	F2 + 900 kHz and F2 + 1 800 kHz	10 kHz
6, 8, 13	-40 dBm	F1 - 1,25 MHz and F1 - 2,50 MHz	10 kHz
	-40 dBm	F2 + 1,25 MHz and F2 + 2,50 MHz	10 kHz
NOTE: F1 = Centre frequency of the lowest supported CDMA channel in the pass band under test. F2 = Centre frequency of the highest supported CDMA channel in the pass band under test.			

Table 4.2.5.2-2: Input intermodulation requirements for interfering signals in the GSM900 and DCS1800 bands

Deployment Environment	Interfering CW signal levels	Frequencies of interfering CW signals	Measurement bandwidth
Co-existence with GSM900 and/or DCS1800 in same geographic area	-15 dBm	876 MHz to 915 MHz 921 MHZ to 960 MHz	10 kHz
	-15 dBm	1 710 MHZ to 1 785 MHz 1 805 MHZ to 1 880 MHz	10 kHz

For the parameters specified in tables 4.2.5.2-1 or 4.2.5.2-2, the power in the operating band shall not increase with more than the limit in table 4.2.5.2-3 at the output of the Repeater as measured in the centre of the operating band, compared to the level obtained without interfering signals applied.

Table 4.2.5.2-3: General input intermodulations limit

Limit for the increase of power in the operating band
+10 dB

4.2.5.3 Conformance

Conformance tests described in clause 5.3.4 shall be carried out.

4.2.6 Out-of-band gain

4.2.6.1 Definition

In the intended application of a Repeater, the out-of-band gain of the Repeater must be less than the coupling loss to the donor Base Station in order to ensure that emissions from the Base Station are not amplified to levels that exceed emissions limits.

4.2.6.2 Limits

The intended use of a Repeater in a system is to amplify the in band signals and not to amplify the out-of-band emission of the donor Base Station.

The manufacturer shall provide information that specifies the minimum donor coupling loss as a function of the Repeater gain setting, and Repeater pass band.

The measured out-of-band gain in the forward link signal path shall be below the minimum donor coupling loss for the configured gain setting (for Repeaters with more than one pass band, each pass band is evaluated).

The manufacturer shall declare the pass band of the Repeater, which shall comply with the frequency coverage requirements of TIA-1030 [6].

For pass bands in which the maximum measured gain is more than 40 dB, when the Repeater is set to its maximum gain setting, the ratio of the inband gain to the out-of-band gain shall comply with the minimum attenuation limits in table 4.2.6.2-1.

For pass bands in which the maximum measured gain is 40 dB or less, when the Repeater is set to its maximum gain setting, the out-of-band gain shall comply with the maximum gain limits in table 4.2.6.2-1.

Table 4.2.6.2-1: Attenuation and gain requirements

For $\|\Delta f\|$ Within the Range	Minimum Attenuation	Maximum Gain
2,50 MHz to 3,75 MHz	≥ 40 dB	≤ 40 dB
3,75 MHz to 12,50 MHz	≥ 40 dB	≤ 35 dB
12,50 MHz to 50,00 MHz	≥ 40 dB	≤ 30 dB
> 50,00 MHz	≥ 40 dB	≤ -10 dB
NOTE: For pass bands that only support a single channel: Δf = centre frequency of supported channel - measurement frequency (f). For pass bands supporting two or more channels, Δf is defined: - for positive Δf as the centre frequency of the highest supported channel - measurement frequency (f); and - for negative Δf as the centre frequency of the lowest supported channel - measurement frequency (f). The limits shall apply for all values of Δf regardless of whether the measurement frequency falls inside or outside of the band or block.		

4.2.6.3 Conformance

Conformance tests described in clause 5.3.5 shall be carried out.

NOTE: Requirements related to the Adjacent Channel Rejection Ratio are captured already in the above requirements on out-of-band gain.

4.2.7 Output intermodulation

4.2.7.1 Definition

Output intermodulation is a measure of the ability of the Repeater to inhibit the generation of intermodulation product signals created by the presence of an interfering signal reaching the Repeater via an output port. The frequency of the interfering signal be ± one, two and three carrier channel spacing frequency offset from the wanted channel (1,23 MHz or 1,25 MHz between channel centre frequencies depending on the band class under test), but within the band class under test.

This test only applies to the Forward link signal path of the Repeater.

4.2.7.2 Limits

The measured power contained in the intermodulation products shall not exceed the requirements listed in the appropriate section of clauses 4.2.2 and 4.2.3 based upon the band class under test.

4.2.7.3 Conformance

Conformance tests described in clause 5.3.6 shall be carried out.

5 Testing for compliance with technical requirements

5.1 Conditions for testing

5.1.1 Introduction

Tests defined in the present document shall be carried out at representative points within the boundary limits of the declared operational environmental profile.

Where technical performance varies subject to environmental conditions, tests shall be carried out under a sufficient variety of environmental conditions (within the boundary limits of the declared operational environmental profile) to give confidence of compliance for the affected technical requirements.

Normally it should be sufficient for all tests to be conducted using standard test conditions except where otherwise stated (see TIA-1037 [5]). For a definition of standard test conditions and for guidance on the use of other test conditions to be used in order to show compliance reference can be made to annex C.

5.1.2 Standard equipment under test

The equipment under test shall be assembled and any necessary adjustments shall be made in accordance with the manufacturer's instructions for the mode of operation required. When alternative modes are available, the equipment shall be assembled and adjusted in accordance with the relevant instructions. A complete series of measurements shall be made for each mode of operation.

5.2 Interpretation of the measurement results

The interpretation of the results recorded in a test report for the measurements described in the present document shall be as follows:

- the measured value related to the corresponding limit shall be used to decide whether an equipment meets the requirements of the present document;
- the value of the measurement uncertainty or the accuracy of each piece of test equipment used for the measurement of each parameter shall be included in the test report; only test equipment meeting the performance requirements for standard test equipment as defined in ANSI/TIA-97 [1], clause 6.4 or 3GPP2 C.S0032-B [4], clause 6.4, shall be used;
- the recorded value of the measurement uncertainty or the recorded value of the accuracy of each piece of test equipment shall be equal to or better than the figures in ANSI/TIA-97 [1], clause 6.4 or 3GPP2 C.S0032-B [4], clause 6.4.

5.3 Essential radio test suites

5.3.1 Spectrum emission mask

5.3.1.1 Initial conditions

Test environment: Normal: see clause 5.1 in TIA-1037 [5].

1) Connect the equipment as shown in figure 6.1-9 in TIA-1037 [5] for the forward link signal path.

2) Connect one signal generator to the input port of the Repeater for tests of Repeaters with a pass band corresponding to one carrier.
If the Repeaters pass band under test is capable of supporting two or more carriers, two signal generators with a combining circuit or one signal generator with the ability to generate two CDMA carriers is connected to the input.

3) The 1 MHz measurement bandwidth may be calculated by integrating multiple 50 kHz or narrower filter measurements.

5.3.1.2 Procedures

1) Configure the Repeater to its maximum rated gain. If the Repeater supports an adjustable pass band span or adjustable supported channels, see clause 5.2 in TIA-1037 [5] for additional test requirements.

2) Configure the CDMA signal generator(s) at the Repeater input port to generate signal(s) in accordance to CDMA test signal, clause 6.2 in TIA-1037 [5], at level(s) that produces the manufacturer specified maximum output power specified in clause 5.2.5 in TIA-1037 [5], at maximum gain.

3) Measure the mean spurious emission levels with ten or more averages and note that the measured value does not exceed the specified value.

4) Increase the input signal level by 10 dB compare to the level obtained in step 2).

5) Measure the mean output power level of the carrier with ten or more averages.

6) Measure the mean spurious emission levels with ten or more averages and note that the measured value does not exceed the specified value.

7) Repeat steps 2) through 6) for each pass band of the Repeater.

8) Repeat steps 2) through 7) for the reverse link signal path of the Repeater.

9) Configure the Repeater to its minimum gain setting and repeat steps 2) through 8).

If the pass band corresponds to two or more consecutive nominal carrier, the requirement shall be met with any combination of two CDMA modulated signals of equal power in the Repeaters pass band.

The results obtained shall be compared to the limits in clause 4.2.2.2 in order to prove compliance.

5.3.2 Spurious emissions

5.3.2.1 Initial conditions

The initial conditions for testing spurious emissions should follow the procedure outlined in clause 5.3.1.1, initial conditions for testing spectrum emission mask.

5.3.2.2 Procedure

The procedure for testing spurious emissions should follow the procedure outlined in clause 5.3.1.2, procedure for testing spectrum emission mask.

The results obtained shall be compared to the limits in clause 4.2.3.2 in order to prove compliance.

5.3.3 Maximum output power

5.3.3.1 Initial conditions

Test environment: Normal: see clause 5.1 in TIA-1037 [5]; and
Extreme: see clause 4.1 in TIA-1037 [5].

1) Connect the equipment as shown figure 6.1-7, clause 6 in TIA-1037 [5] for the Forward link signal path.

5.3.3.2 Procedure

1) Configure the Repeater to its maximum rated gain. If the Repeater supports an adjustable pass band span or adjustable supported channels, see clause 5.2 in TIA-1037 [5] for additional test requirements.

2) Configure the Repeater to transmit a single carrier in one of its pass bands. This is the pass band under test.

3) Configure the CDMA signal generator at the Repeater input port to generate a signal in accordance to CDMA test signal, clause 6.2 in TIA-1037 [5] at a level that produces the manufacturer specified maximum output power specified in clause 5.2.5 in TIA-1037 [5], at maximum gain. Locate at the CDMA channel nearest the centre of the pass band under test.

4) Measure the mean output power level of the carrier with ten or more averages and note that the measured value does not exceed the specified value.

5) Increase the input signal to the Repeater by a total of 10 dB.

6) Measure the mean output power level of the carrier with ten or more averages at least 3 seconds after performing step 4).

7) Repeat steps 1) through 6) for each pass band of the Repeater.

8) Repeat steps 1) through 7) for the reverse link signal path of the Repeater.

9) Repeat steps 1) through 8), using the extreme test environment requirements and method contained in clause 4.1 in TIA-1037 [5].

The results obtained shall be compared to the limits in clause 4.2.4.2 in order to prove compliance.

5.3.4 Input intermodulation

5.3.4.1 Initial conditions

Test environment: Normal: see clause 5.1 in TIA-1037 [5].

1) Connect the equipment as shown in figure 6.1-4 in TIA-1037 [5] for the Forward link signal path.

5.3.4.2 Procedures

1) Configure the Repeater to its maximum gain setting. If the Repeater supports adjustable pass band span or adjustable supported channels, see clause 5.2 in TIA-1037 [5] for additional test requirements.

2) For the Band Class and Band Subclass under test perform steps 3) through 11).

3) Adjust the CW generators as required to obtain the input power level listed in table 4.2.5.2-1 for the F1 test.

4) Measure the mean output power of the Repeater with ten or more averages by integrating the power contained in the lowest supported CDMA channel in the pass band under test (centred about F1), using a 10 kHz RBW.

5) Remove the signal generators from the input of the Repeater and terminate the input with 50 Ω.

6) Measure the mean output power of the Repeater with ten or more averages by integrating the power contained in the lowest supported CDMA channel in the pass band under test (centred about F1), using a 10 kHz RBW.

7) Adjust the CW generators as required to obtain the input power level listed in table 4.2.5.2-1 for the F2 test.

8) Measure the mean output power of the Repeater with ten or more averages by integrating the power contained the highest supported CDMA channel in the pass band under test (centred about F2), using a 10 kHz RBW.

9) Remove the signal generators from the input of the Repeater and terminate the input with 50 Ω.

10) Measure the mean output power of the Repeater with ten or more averages by integrating the power contained in the highest supported CDMA channel in the pass band under test (centred about F2), using a 10 kHz RBW.

11) If the manufacturer has specified that the Repeater is not suitable for deployment in a co-existence or co-location environment with GSM900 and/or DCS1800, skip to step 17). If the pass band under test is declared to support CDMA channel numbers that would contain the interfering frequencies listed in table 4.2.5.2-2, skip to step 16). If neither of these exclusions is valid, perform steps 12) through 15).

12) Remove the 50 Ω termination from the Repeater input and re-connect the signal generators. Apply two CW signals at the levels and frequencies as indicated in table 4.2.5.2-2 such that the lowest order intermodulation products are located in the centre of the pass band under test (so that both intermodulation products are within ±100 kHz of the centre of the pass band under test).

13) Measure the mean output power of the Repeater with ten or more averages by integrating the power contained in the CDMA channel located nearest to the centre of the pass band under test, using a 10 kHz RBW.

14) Remove the signal generators from the input of the Repeater and terminate the input with 50 Ω.

15) Measure the mean output power of the Repeater with ten or more averages by integrating the power contained in the CDMA channel located nearest to the centre of the pass band under test, using a 10 kHz RBW.

16) Repeat steps 1) through 15) for each pass band supported by the Repeater.

17) Repeat steps 1) through 16) for the reverse link signal path of the Repeater.

The results obtained shall be compared to the limits in clause 4.2.5.2 in order to prove compliance.

5.3.5 Out-of-band gain

5.3.5.1 Initial conditions

Test environment: Normal: see clause 5.1 in TIA-1037 [5].

1) Connect the equipment as shown in figure 6.1-1 in TIA-1037 [5] for the Forward link signal path.

5.3.5.2 Procedure

1) Configure the Repeater to its maximum gain setting. If the Repeater supports an adjustable pass band span or adjustable supported channels, see clause 5.2 in TIA-1037 [5] for additional test requirements.

2) Configure the signal generator at the Repeater input port to produce a CW signal at an amplitude approximately 5 dB below the level that would cause the maximum rated mean output power from the Repeater for the configured gain setting at the centre of its pass band.

3) For pass bands in which the maximum measured gain is more than 40 dB, when the Repeater is set to its maximum gain setting:
Measure the minimum attenuation relative to the minimum gain in the pass band under test at frequency offsets from the highest supported CDMA channel centre frequency, as detailed in TIA-1037 [5].

4) For pass bands in which the maximum measured gain is more than 40 dB, when the Repeater is set to its maximum gain setting:
Measure the minimum attenuation relative to the minimum gain in the pass band under test at frequency offsets from the lowest supported CDMA channel centre frequency, as detailed in table 4.2.6.2-1.

5) For pass bands in which the maximum measured gain is 40 dB or less, when the Repeater is set to its maximum gain setting:
Measure the maximum gain at frequency offsets from the highest supported CDMA channel centre frequency in the pass band under test, as detailed in table 4.2.6.2-1.

6) For pass bands in which the maximum measured gain is 40 dB or less, when the Repeater is set to its maximum gain setting:
Measure the maximum gain at frequency offsets from the lowest supported CDMA channel centre frequency in the pass band under test, as detailed in table 4.2.6.2-1.

7) Repeat steps 2) through 6) for each pass band supported by the Repeater. (Each pass band is independently evaluated; i.e. gain from a pass band that is not under test is ignored.)

8) Repeat steps 2) through 7) for the reverse link signal path of the Repeater.

The results obtained shall be compared to the limits in clause 4.2.6.2.

5.3.6 Output intermodulation

5.3.6.1 Initial conditions

Test environment: Normal: see clause 5.1 in TIA-1037 [5].

1) Connect the equipment as shown in figure 6.1-8 in TIA-1037 [5] for the Forward link signal path.

5.3.6.2 Procedure

1) Configure the Repeater to its maximum gain setting. If the Repeater supports an adjustable pass band span, adjustable supported channels, see clause 5.2 in TIA-1037 [5] for additional test requirements.

2) Connect a signal generator to the circulator on the input port of the Repeater. Connect another signal generator to the circulator on the output port and make sure the signal generator's power is directed to the Repeater output port.

3) Configure this CDMA signal generator at the Repeater (subject signal) input port to generate a signal in accordance to CDMA test signal, clause 6.2 in TIA-1037 [5], at a level that produces the manufacturer's specified maximum output power from the Repeater, at the CDMA channel nearest the centre of the pass band under test.

4) Configure the signal generator on the circulator connected to the output port of the Repeater (interfering signal) to generate a signal in accordance to CDMA test signal, clause 6.2 in TIA-1037 [5], at a level that produces a signal power corresponding to 30 dB below the manufacturer's specified maximum output power of the Repeater with a frequency offset of one carrier channel spacing (1,23 MHz or 1,25 MHz between channel centre frequencies depending on the band class under test) above the Repeater output frequency.

5) Measure the emission at the third and fifth order intermodulation products of the Repeater output signal and the interfering signal using a true RMS detector and the bandwidth specified for the frequency offset based on the band class under test as specified in the appropriate section of clauses 4.2.2 and 4.2.3.

6) Configure the signal generator on the circulator connected to the output port of the Repeater (interfering signal) to generate a CDMA signal (see clause 6.2 in TIA-1037 [5]) that produces a signal power corresponding to 30 dB below the manufacturer's specified maximum mean output power of the Repeater with a frequency offset of one carrier channel spacing (1,23 MHz or 1,25 MHz between channel centre frequencies depending on the band class under test) below the Repeater output frequency.

7) Measure the emission at the third and fifth order intermodulation products of the Repeater output signal and the interfering signal using a true RMS detector and the bandwidth specified for the frequency offset based on the band class under test as specified in the appropriate section of clauses 4.2.2 and 4.2.3.

8) Repeat steps 4) through 7) using frequency offsets of 2 times the carrier channel spacing.

9) Repeat steps 4) through 7) using frequency offsets of 3 times the carrier channel spacing.

10) Repeat steps 3) through 9) for each Forward link pass band supported by the Repeater.

The results obtained shall be compared to the limits in clause 4.2.7.2 in order to prove compliance.

Annex A (normative): HS Requirements and conformance Test specifications Table (HS-RTT)

The HS Requirements and conformance Test specifications Table (HS-RTT) in table A-1 serves a number of purposes, as follows:

- it provides a statement of all the requirements in words and by cross reference to (a) specific clause(s) in the present document or to (a) specific clause(s) in (a) specific referenced document(s);
- it provides a statement of all the test procedures corresponding to those requirements by cross reference to (a) specific clause(s) in the present document or to (a) specific clause(s) in (a) specific referenced document(s);
- it qualifies each requirement to be either:
  - Unconditional: meaning that the requirement applies in all circumstances; or
  - Conditional: meaning that the requirement is dependent on the manufacturer having chosen to support optional functionality defined within the schedule;
- in the case of Conditional requirements, it associates the requirement with the particular optional service or functionality;
- it qualifies each test procedure to be either:
  - Essential: meaning that it is included with the Essential Radio Test Suite and therefore the requirement shall be demonstrated to be met in accordance with the referenced procedures;
  - Other: meaning that the test procedure is illustrative but other means of demonstrating compliance with the requirement are permitted.

Table A-1: HS Requirements and conformance Test specifications Table (HS-RTT)

Harmonized Standard EN 301 908-12 The following requirements and test specifications are relevant to the presumption of conformity under the article 3.2 of the R&TTE Directive						
Requirement			**Requirement Conditionality**		**Test Specification**	
No	**Description**	**Reference: Clause No**	**U/C**	**Condition**	**E/O**	**Reference: Clause No**
1	Spectrum emissions mask	4.2.2	U		E	5.3.1
2	Spurious emissions	4.2.3	U		E	5.3.2
3	Maximum output power	4.2.4	U		E	5.3.3
4	Input intermodulation	4.2.5	U		E	5.3.4
5	Out-of-band gain	4.2.6	U		E	5.3.5
6	Output intermodulation	4.2.7	U		E	5.3.6

Key to columns:

Requirement:

No A unique identifier for one row of the table which may be used to identify a requirement or its test specification.

Description A textual reference to the requirement.

Clause Number Identification of clause(s) defining the requirement in the present document unless another document is referenced explicitly.

Requirement Conditionality:

U/C Indicates whether the requirement is to be *unconditionally* applicable (U) or is *conditional* upon the manufacturers claimed functionality of the equipment (C).

Condition Explains the conditions when the requirement shall or shall not be applicable for a technical requirement which is classified "conditional".

Test Specification:

E/O Indicates whether the test specification forms part of the Essential Radio Test Suite (E) or whether it is one of the Other Test Suite (O).

NOTE: All tests whether "E" or "O" are relevant to the requirements. Rows designated "E" collectively make up the Essential Radio Test Suite; those designated "O" make up the Other Test Suite; for those designated "X" there is no test specified corresponding to the requirement. The completion of all tests classified "E" as specified with satisfactory outcomes is a necessary condition for a presumption of conformity. Compliance with requirements associated with tests classified "O" or "X" is a necessary condition for presumption of conformity, although conformance with the requirement may be claimed by an equivalent test or by manufacturer's assertion supported by appropriate entries in the technical construction file.

Clause Number Identification of clause(s) defining the test specification in the present document unless another document is referenced explicitly. Where no test is specified (that is, where the previous field is "X") this field remains blank.

Annex B (normative): Repeater configurations

B.1 Power supply

When extreme power supply conditions are specified for a test, the test shall be performed at the standard upper and lower limits of operating voltage defined by manufacturer's declaration for the equipment under test.

B.2 Power supply options

If the Repeater is supplied with a number of different power supply configurations, it may not be necessary to test RF parameters for each of the power supply options, provided that it can be demonstrated that the range of conditions over which the equipment is tested is at least as great as the range of conditions due to any of the power supply configurations.

B.3 Ancillary equipment

The Repeater equipment may include ancillary equipment during tests, if the ancillary equipment is normally used in the operation of the equipment under test. This would include power supplies, cabinets, antenna couplers, and receiver multi-couplers.

Annex C (informative): Environmental profile specification

The following environmental conditions may be declared by the supplier:

- barometric pressure: minimum and maximum;
- temperature: minimum and maximum;
- relative humidity: minimum and maximum;
- power supply: lower and upper voltage limit.

When operating outside the boundary limits of the declared operational environmental profile the equipment should not make ineffective use of the radio frequency spectrum so as to cause harmful interference.

Annex D (informative): The EN title in the official languages

The enlargement of the European Union (EU) resulted in a requirement from the EU for a larger number of languages for the translation of the titles of Harmonized Standards and mandated ENs that are to be listed in the Official Journal to support the implementation of this legislation.

For this reason the title translation concerning the present document can be consulted via the e-approval application.

Annex E (informative): Bibliography

ETSI TR 100 028 (V1.4.1): "Electromagnetic Compatibility and Radio Spectrum Matters (ERM); Uncertainties in the measurement of mobile radio equipment characteristics".

ETSI EN 301 489 (series): "Electromagnetic compatibility and Radio spectrum Matters (ERM); ElectroMagnetic Compatibility (EMC) standard for radio equipment and services".

Directive 2004/108/EC of the European Parliament and of the Council of 15 December 2004 on the approximation of the laws of the Member States relating to electromagnetic compatibility and repealing Directive 89/336/EEC (text with EEA relevance (EMC Directive).

Council Directive 73/23/EEC of 19 February 1973 on the harmonization of the laws of Member States relating to electrical equipment designed for use within certain voltage limits (LV Directive).

CEPT/ERC/REC 74-01E (Siófok 1998, Nice 1999, Sesimbra 2002, Hradec Kralove 2005): "Unwanted Emissions in the Spurious Domain".

History

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